FEDERATED LEARNING - BASED INTRUSION DETECTION SYSTEM FOR THE INTERNET OF MEDICAL THINGS (IoMT)

A PROJECT REPORT

Submitted by

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BONAFIDE CERTIFICATE

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ABSTRACT

The Internet of Medical Things (IoMT) has enhanced patient care through connected medical devices but has also increased cybersecurity risks, making intrusion detection essential for patient safety. Conventional Intrusion Detection Systems (IDS) often rely on centralized data processing, raising privacy concerns and creating potential security vulnerabilities.This project proposes a federated, explainable, and blockchain-audited IDS for multi-hospital IoMT environments. Network traffic is processed locally within each hospital, where an XGBoost model detects potential attacks. Using the Flower framework, only model updates are shared with a central server through Federated Averaging (FedAvg), ensuring that raw clinical data remains within each hospital. A risk-adaptive scoring mechanism evaluates attack probability and device trust, while SHAP provides feature-level explanations for detected threats. A SHA-256 hash-chained ledger maintains tamper-evident records of alerts and model updates. The system integrates a React.js dashboard, FastAPI backend, and PostgreSQL database for monitoring and management. The proposed framework is designed for evaluation using the ECU-IoHT and WUSTL-EHMS-2020 datasets. By integrating privacy preservation, explainability, and auditability, the system aims to improve secure and accountable intrusion detection in healthcare networks. Additionally, the system aims to support early identification of suspicious activities and prioritize alerts based on their risk levels. The combination of federated learning and local processing reduces the need for centralized storage of sensitive healthcare data. Explainable predictions help security teams understand the reasons behind detected attacks and take appropriate action.

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CHAPTER 1
INTRODUCTION

1.1 BACKGROUND

The Internet of Medical Things (IoMT) has transformed healthcare by enabling continuous monitoring through connected medical devices such as heart-rate monitors, infusion pumps, and wearable sensors. However, this connectivity introduces cybersecurity risks due to the limited computing and memory resources of many IoMT devices. A compromised device can expose the hospital network to attacks such as man-in-the-middle spoofing, data injection, and denial-of-service, potentially affecting patient safety and clinical decisions. Most conventional intrusion detection systems (IDS) rely on centralized architectures that collect network and device data from multiple hospitals for training and inference. This approach raises privacy and data-protection concerns, as hospitals are often reluctant to share sensitive patient-linked information. Furthermore, centralized servers create a single point of failure and an attractive target for attackers.

1.2 PROBLEM STATEMENT

Existing IoMT intrusion detection systems are largely built around a centralized data-collection model that requires raw traffic and, in some cases, patient biometric data to be transmitted to and stored at a central server for model training and inference. This creates three compounding problems for hospital deployments: first, it conflicts with data-protection regulation and hospital privacy policy, since patient-linked network traffic should not leave the facility 1 that generated it; second, it creates a single point of failure, since a compromise of the central server exposes the training data and detection logic of every participating hospital at once; and third, most existing systems return only a binary "attack" or "normal" label with little or no explanation, which gives hospital security staff no way to judge whether an alert deserves urgent attention or to trust the system's decision at all. There is a need for an intrusion detection architecture that keeps hospital data local, produces prioritized and explainable alerts, and maintains a verifiable record of its own decisions and model updates.

1.3 OBJECTIVES

- To design a federated intrusion detection architecture for IoMT in which raw patient and network data remain within each hospital and only model updates are exchanged with a central coordinating server.
- To use the Flower federated learning framework to coordinate Federated Averaging (FedAvg) across per-hospital XGBoost classifiers, producing a global detection model without centralizing raw data.

- To build a risk-adaptive detection layer that converts raw model output into an attack-probability score, a per-device trust score, and a discrete risk classification, rather than a single binary label.
- To integrate SHAP-based explainability directly into the detection pipeline so that every alert carries a feature-level explanation of why it was raised.
- To implement a blockchain-inspired, SHA-256 hash-chained audit layer that provides a tamper-evident record of alerts and federated model updates.
- To expose the complete pipeline through a React.js security dashboard backed by a FastAPI service layer and a PostgreSQL database, and to evaluate the design on the ECU-IoHT and WUSTL-EHMS-2020 datasets.

1.4 SCOPE

The scope of this project covers the design and staged implementation of a privacy-preserving intrusion detection system for multi-hospital IoMT 2 deployments. It includes the edge-side capture and pre-processing of IoMT network traffic, per-hospital local training of XGBoost classifiers, coordination of federated training rounds through the Flower framework, a risk-adaptive alert scoring layer, a SHAP-based explainability module, a SHA-256 hash-chained audit ledger, and a web-based security dashboard. The system is scoped to two representative public datasets, ECU-IoHT and WUSTL-EHMS-2020, which respectively provide device-level attack traffic and combined network-biometric attack traffic for IoMT environments. Deployment concerns beyond the prototype stage, such as integration with a hospital's existing electronic health record system or certification against medical-device regulation, are outside the scope of the present work but are noted as directions for future development.

The implementation also covers monitoring attack behaviour, generating risk-based alerts, and maintaining audit records for security analysis. The dashboard provides a centralized view of detection results without requiring hospitals to share their raw clinical data. Deployment concerns beyond the prototype stage, such as integration with a hospital's existing electronic health record system or certification against medical-device regulation, are outside the scope of the present work but are noted as directions for future development.

1.5 SECURITY REQUIREMENTS

The system follows the Confidentiality, Integrity, and Availability (CIA) triad, adapted for federated, multi-hospital IoMT environments. Confidentiality is ensured by keeping raw network traffic and patient biometric data within each hospital, sharing only model updates for federated aggregation. Integrity is supported through resilient aggregation against corrupted updates and a SHA-256 hash-chained ledger that detects tampering with alerts and model-update records. Availability is maintained by performing detection locally at hospital edge gateways, ensuring that network outages or central server attacks do not interrupt in-hospital detection. The system also treats IoMT devices as untrusted endpoints and monitors their network traffic for potential compromises. Additionally, secure communication is considered to protect model updates during

transmission between participating hospitals and the federated server. Access to the security dashboard and sensitive records is restricted to authorized users. SHAP-based explanations support transparent analysis of detected threats, while regular verification of the audit ledger helps maintain the reliability of stored security records. These considerations aim to protect patient privacy, preserve the integrity of detection results, and support continuous monitoring of healthcare networks.

1.6 SDG ALIGNMENT

The Project contributes to the United Nations Sustainable Development Goals through its focus on protecting the digital infrastructure that underpins modern healthcare delivery.

Table 1.1 SDG Alignment of the Project

SDG NO. & GOAL	TITLE	RELEVANT TARGET	PROJECT ALIGNMENT
SDG 3	Good Health and Well-Being	Ensure healthy lives and promote wellbeing	Protects the integrity of IoMT devices and the vital-sign data clinicians rely on, reducing the risk that a cyberattack disrupts patient monitoring or treatment.
SDG 9	Industry, Innovation and Infrastructure	Build resilient infrastructure and foster innovation	Introduces a federated, explainable, and auditable security architecture that strengthens the resilience of connected healthcare infrastructure.

SDG 11	Sustainable Cities and Communities	Strengthen efforts to protect cultural and natural heritage build resilient comunities	Supports safer, technology-enabled healthcare facilities as part of resilient urban and community infrastructure.

CHAPTER 2

LITERATURE REVIEW

2.1 LITERATURE SURVEY

Table 2.1 Comparison With Representative Existing Systems

S.No.	Author(s)	Proposed Approach	Key Findings
1	A. Ghourabi and A. Alkhalil	Federated learning model for detecting cyberattacks in IoMT networks	Applies federated learning to enable collaborative and privacy-preserving cyberattack detection in IoMT networks.
2	FTL-TSLP	Federated transfer learning with a two-stage LSTM pipeline for IoMT intrusion detection	Provides privacy-preserving and fault-tolerant intrusion detection for IoMT networks.
3	A. Berguiga, A. Harchay, and A. Massaoudi	HIDS-IoMT: Deep-learning-based intelligent intrusion detection system for IoMT	Uses deep learning to detect attacks and improve intelligent intrusion detection in IoMT environments.
4	XAI-EdgeSFL	Explainable split federated learning with adaptive intrusion-resilient edge intelligence for healthcare	Combines federated learning, edge intelligence, intrusion resilience, and explainability for healthcare ecosystems.
5	V. T. Nguyen and R. Beuran	FedMSE: Semi-supervised federated learning for IoT network	Uses semi-supervised federated learning to perform intrusion

		intrusion detection	detection while reducing dependence on labelled data.
6	Z. Li, W. Yao, J. Luo, and Z. Huang	Flow-based IoT intrusion detection using improved generative federated distillation learning	Uses federated distillation to support collaborative flow-based intrusion detection while preserving distributed data.
7	M. Baich and N. Sael	Federated learning-based intrusion detection using dynamic ensemble aggregation for IoT networks	Uses dynamic ensemble aggregation to improve federated intrusion detection in distributed IoT environments.

2.2 RESEARCH GAPS

The literature reviewed above shows that federated learning, blockchain-based integrity protection, explainable AI, and trust-aware detection have each been demonstrated to work well individually, and combinations of any two of them appear in recent work. The following gaps remain and motivate the design proposed in this project:

- Few systems combine federated training, blockchain-backed audit, and explainability within a single deployable pipeline; most works address only one or two of these concerns at a time.
- Most reported systems emit a flat binary label for a detected event rather than a graded, risk-adaptive alert that a hospital security team could realistically triage under alert fatigue.
- Device-level trust is rarely modelled explicitly; a device's recent behavioural history is generally not incorporated into the final risk score alongside the raw classifier output.
- Evaluation is often limited to a single dataset or attack modality, leaving it unclear whether a proposed system generalizes across both device/network-level attacks (as in ECU-IoHT) and combined network biometric attacks (as in WUSTL-EHMS-2020).
- Few works present a complete, operator-facing dashboard that consolidates alerts, explanations, federated training status, and audit-trail verification in one place.

2.3 PROPOSED SYSTEM

2.3.1 Overview

The system proposed in this project is a federated, explainable, and blockchain audited intrusion detection pipeline built specifically for multi hospital IoMT deployments. Traffic from IoMT devices is captured and pre processed at an edge gateway inside each hospital; a local XGBoost classifier is trained on this data without it ever leaving the site. Only model weights are shared with a Flowercoordinated federated learning server, which aggregates them using Federated Averaging to produce a global detection model. Detected events pass through a risk-adaptive scoring layer that produces an attack probability score, a device trust score, and a discrete risk classification, and every prediction is explained using SHAP. Alerts and model-update hashes are written to a SHA-256 hashchained ledger, and the complete pipeline is exposed through a React.js dashboard backed by FastAPI and PostgreSQL. The proposed system enables hospitals to collaboratively improve attack detection while maintaining the privacy of sensitive clinical information. It supports early identification of suspicious activities, provides understandable explanations for security alerts, and maintains reliable audit records for future verification. The overall design aims to improve security monitoring, transparency, and accountability in healthcare networks while reducing dependence on centralized data storage.

2.3.2 Objectives

- Preserve hospital data privacy by design, so that no raw traffic or patient data is transmitted outside the originating hospital at any stage of training or inference.
- Provide graded, prioritized alerts through the risk-adaptive scoring layer rather than a flat binary classification.
- Make every detection decision explainable to a non-specialist security analyst through SHAP-based feature attribution.
- Guarantee that the historical record of alerts and model updates cannot be silently altered, through a hash-chained audit ledger.
- Present all of the above through a single, operationally usable dashboard.
- Support collaborative learning among hospitals to improve the global model using locally trained updates.

2.3.3 Advantages

- Keeps sensitive hospital data local, reducing regulatory exposure and removing the central data-pooling single point of failure found in most existing IoMT-IDS designs.
- Combines federated learning, explainability, and blockchain-backed auditability within one pipeline rather than treating them as independent add-ons.
- Converts raw predictions into an actionable, risk-graded alert, which is closer to how a real hospital security operations team would want to consume detection output.
- Uses a lightweight, tree-based classifier (XGBoost) that is inexpensive enough to train at the edge and to explain in near real time with SHAP's TreeExplainer.

- Is evaluated across two complementary public datasets, ECU-IoHT and WUSTL-EHMS-2020, giving broader evidence of generalization than a singledataset evaluation would.
- Supports local detection even during temporary communication failures with the central federated server, reducing dependence on continuous connectivity.
- Improves transparency and accountability by maintaining verifiable security records and providing understandable explanations for detected threats.

CHAPTER 3

SYSTEM DESIGN

3.1 SYSTEM ARCHITECTURE

The architecture of the proposed system follows an edge-to-cloud pipeline organized into eight stages, moving from raw IoMT device traffic to an operatorfacing dashboard, as shown in Figure. 3.1. Traffic generated by IoMT devices is captured and processed at an edge gateway inside each hospital; each hospital then trains a local XGBoost model on its own data and sends only the resulting weights to a central federated learning server, which aggregates them into a global model. The global model's output is passed through a risk-adaptive scoring layer and a SHAP explainability layer, and both the resulting alerts and every federated model update are recorded in a blockchain-style audit ledger. All of this is finally surfaced through a security dashboard.

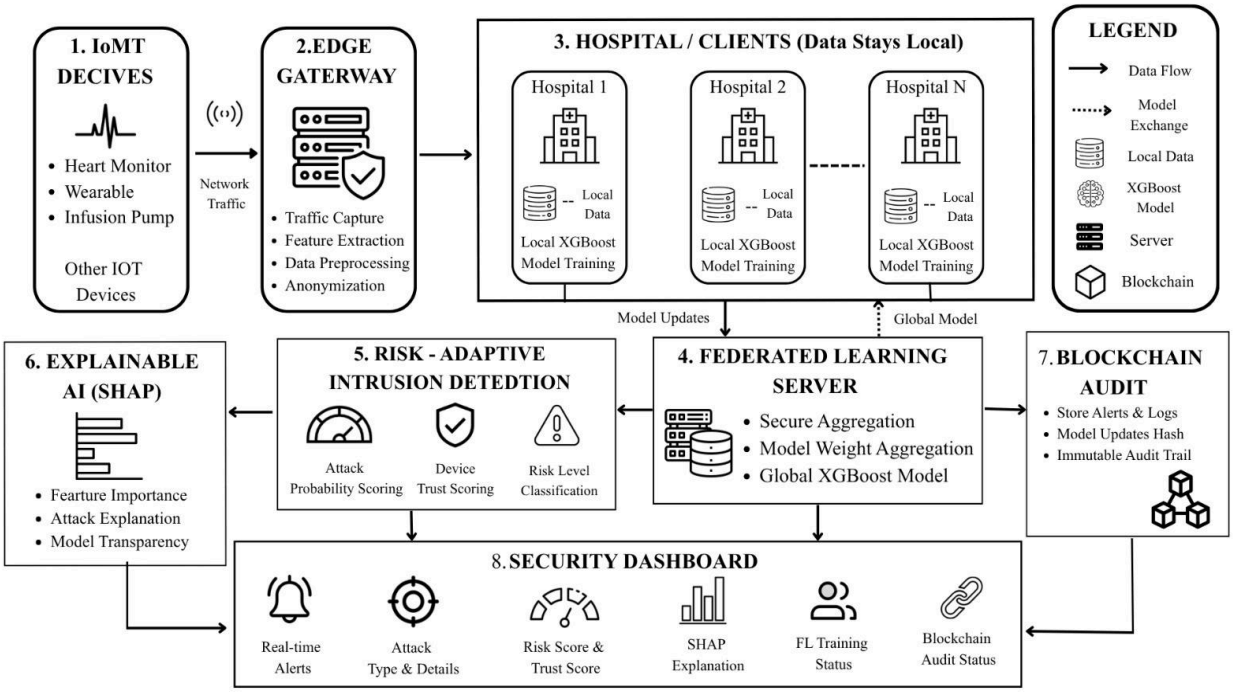


Figure 3.1 Architecture of the Proposed Federated, Explainable, and

Blockchain Audited IDS for IoMT

3.2 MODULE DESIGN

1. IoMT Device Layer

Heart monitors, wearables, infusion pumps, and bedside devices generate continuous patient data and network traffic. These devices are treated as untrusted endpoints, and their traffic is monitored for potential attacks.

2. Edge Gateway Module

The edge gateway captures network traffic within each hospital, extracts statistical and protocol-level features, and performs preprocessing, including normalization and missing-value handling. Sensitive identifying information is anonymized before local training.

3. Hospital-Level Local Training Module

Each hospital trains an XGBoost classifier using its local dataset without sharing raw data. XGBoost is selected for its efficient training on tabular network and biometric features and its compatibility with SHAP-based explanations.

4. Federated Learning Server Module

Hospitals share only model updates with the central server, not raw patient data. Using the Flower framework, the server performs Federated Averaging (FedAvg) to combine updates and distributes the improved global model to participating hospitals.

5. Risk-Adaptive Intrusion Detection Module

This module generates an attack-probability score, a device trust score based on recent behaviour, and a risk classification. These outputs help identify suspicious traffic and prioritize potential threats.

6. Explainable AI Module

The SHAP module explains each prediction by identifying and ranking the network and biometric features that influence the XGBoost model's decision, helping security teams understand detected threats.

7. Blockchain Audit Module

Alerts and federated model-update records are stored in a SHA-256 hash-chained ledger. Each block references the previous block's hash, making unauthorized changes to audit records detectable.

8. Security Dashboard Module

The React.js dashboard displays real-time alerts, attack details, risk and trust scores, SHAP explanations, federated training status, and blockchain audit verification. The dashboard is supported by FastAPI and PostgreSQL for backend services and data storage.

3.3 DATA FLOW DIAGRAMS

The federated learning process is coordinated using the Flower framework, which supports XGBoost through a client-server architecture. Figure 3.2 illustrates one federated learning round. At the beginning of round t, the server distributes the global model to participating hospitals. Each hospital trains the model locally on its own dataset without sharing raw data and sends only the updated parameters to the server. The server aggregates these updates using Federated Averaging (FedAvg), weighted by the number of local training samples, and redistributes the updated global model to all hospitals for the next round.

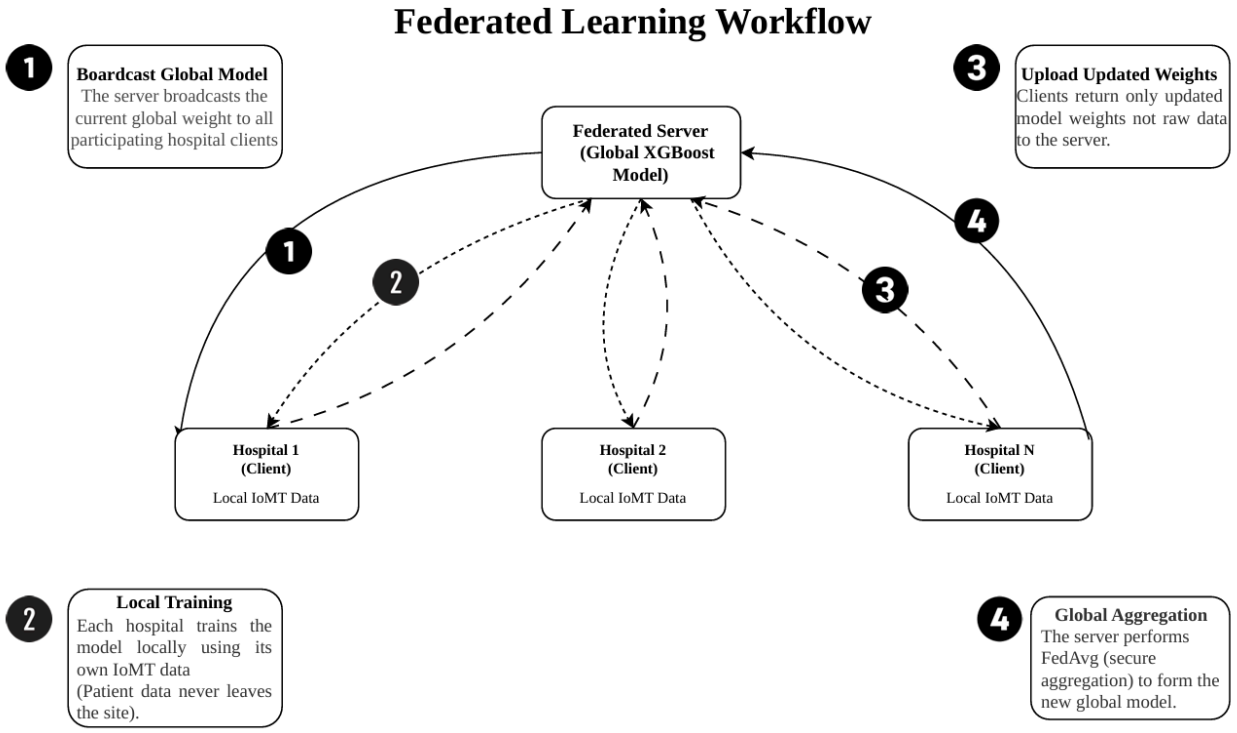


Figure 3.2 Federated Learning Round between the Flower Server and Hospital Clients

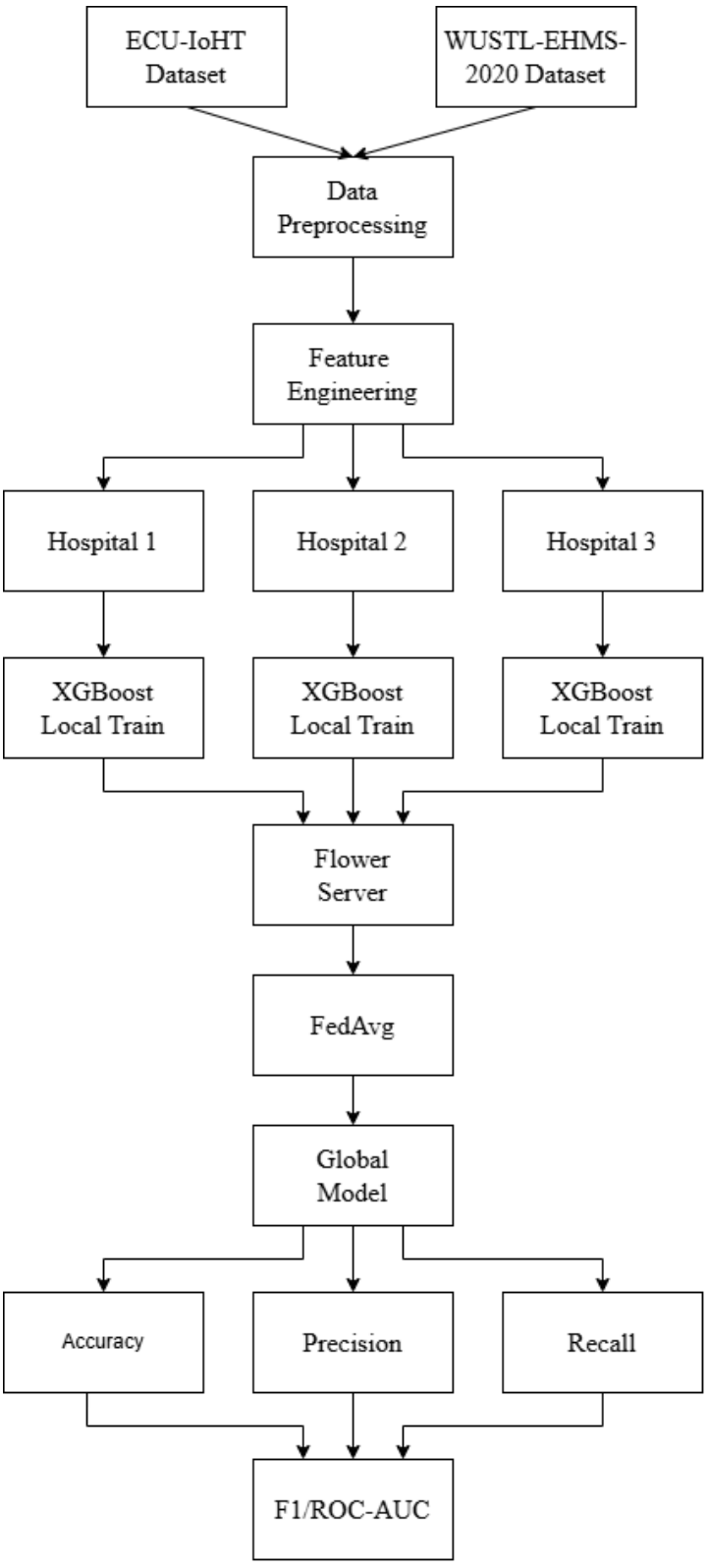


Figure 3.3 Overall implementation and data processing workflow

3.4 THREAT MODEL AND ATTACK SURFACE

The proposed system considers the following attack vectors, their targets, and potential impacts on the multi-hospital IoMT environment.

1. Man-in-the-Middle Spoofing:

An attacker intercepts and monitors communication between IoMT devices, edge gateways, or monitoring servers. This targets the edge gateway and network links, potentially exposing sensitive patient telemetry and affecting data confidentiality.

2. Data Injection:

An attacker modifies packets during transmission to alter vital-sign readings or network-flow values. This targets the edge gateway and network links, potentially corrupting readings, influencing clinical decisions, or helping attacks evade detection.

3. Denial-of-Service on Gateway:

An attacker floods the edge gateway or IoMT network with excessive traffic, disrupting monitoring and detection services. This may affect the availability of real-time monitoring and security alerts.

4. Compromised Hospital Client / Model Poisoning:

A malicious or compromised hospital submits corrupted model updates to the federated learning server. This targets the aggregation process and may reduce global model accuracy or introduce detection weaknesses.

5. Unauthorized Dashboard Access:

An unauthorized user attempts to access alerts, patient-linked metadata, or system configurations through the React.js dashboard or FastAPI backend. This may cause data exposure or unauthorized modification of alert records.

6. Audit Ledger Tampering:

An attacker with backend access attempts to modify stored alerts or model-update records in PostgreSQL or the hash-chained ledger. This threatens the integrity and reliability of the audit trail and may affect accountability.

3.5 SECURITY CONTROLS

1. Man-in-the-Middle Spoofing:

TLS-encrypted communication and mutual authentication are used between IoMT devices, edge gateways, and backend services to protect data during transmission.

2. Data Injection:

Edge-level feature validation, anomaly checks during preprocessing, and per-flow integrity verification help identify modified or suspicious network data.

3. Denial-of-Service on Gateway:

Rate limiting and traffic shaping at the edge gateway help control excessive traffic. Local detection continues even when communication with the federated server is interrupted.

4. Compromised Hospital Client / Model Poisoning:

Federated Averaging is combined with model-update plausibility checks before aggregation to limit the influence of suspicious or outlier updates on the global model.

5. Unauthorized Dashboard Access:

Authentication and role-based access control (RBAC) are enforced through the FastAPI service layer. Clinicians, security analysts, and administrators are provided access according to their assigned roles.

6. Audit Ledger Tampering:

SHA-256 hash chaining is applied to alert and model-update records, with each new block referencing the previous block's hash. A chain-verification process helps detect unauthorized changes to stored audit records.

CHAPTER 4

SYSTEM DESCRIPTION

4.1 DATA COLLECTION

Two complementary public datasets are used to train and evaluate the local and global classifiers, chosen because they represent the two dominant data modalities in IoMT security research: device- and network-level attack traffic, and combined network-biometric traffic.

ECU-IoHT

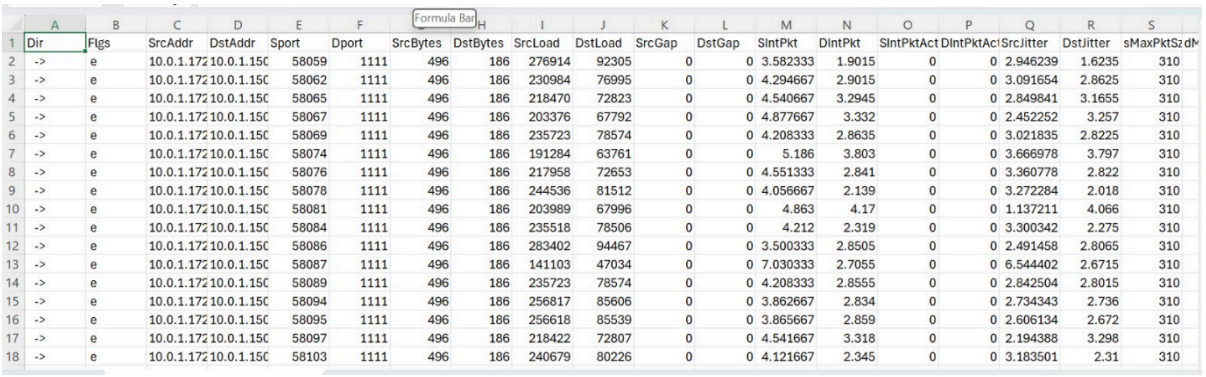
The ECU-IoHT dataset was built by Edith Cowan University on a real Internet of Health Things testbed in which a range of vulnerabilities in connected medical devices were deliberately exploited to generate realistic attack traffic, addressing the shortage of publicly available cybersecurity datasets specific to the healthcare domain. This dataset is used to expose the local classifiers to device- and network-level attack behaviour representative of an IoMT edge segment.

	A	B	C	D	E	F	G	H	I	J
1	No.	Time	Source	Destination	Protocol	Length	Info	Type	Type of attack	
2	1	0	Alfa_97:cf:63	Broadcast	ARP	42	Who has 192.16	Attack	ARP Spoofing	
3	2	0.002956	6e:c7:ec:3c:f2:ba	Alfa_97:cf:63	ARP	42	192.168.43.1 is	Attack	ARP Spoofing	
4	3	0.200725	Alfa_97:cf:63	Broadcast	ARP	42	Who has 192.16	Attack	ARP Spoofing	
5	4	0.202713	192.168.43.186	192.168.43.1	DNS	85	Standard query	Normal	No Attack	
6	5	0.411565	6e:c7:ec:3c:f2:ba	Alfa_97:cf:63	ARP	42	192.168.43.1 is	Attack	ARP Spoofing	
7	6	0.431963	192.168.43.1	192.168.43.186	DNS	144	Standard query	Normal	No Attack	
8	7	0.433064	192.168.43.186	192.168.43.1	TCP	58	36954 > 1720 [	Normal	No Attack	
9	8	0.433084	192.168.43.186	192.168.43.1	TCP	58	36954 > 5900 [	Normal	No Attack	
10	9	0.433089	192.168.43.186	192.168.43.1	TCP	58	36954 > 135 [S	Normal	No Attack	
11	10	0.433092	192.168.43.186	192.168.43.1	TCP	58	36954 > 143 [S	Normal	No Attack	
12	11	0.433096	192.168.43.186	192.168.43.1	TCP	58	36954 > 993 [S	Normal	No Attack	
13	12	0.433099	192.168.43.186	192.168.43.1	TCP	58	36954 > 110 [S	Normal	No Attack	
14	13	0.433103	192.168.43.186	192.168.43.1	TCP	58	36954 > 21 [SY	Normal	No Attack	
15	14	0.433107	192.168.43.186	192.168.43.1	TCP	58	36954 > 113 [S	Normal	No Attack	
16	15	0.433111	192.168.43.186	192.168.43.1	TCP	58	36954 > 22 [SY	Normal	No Attack	
17	16	0.433114	192.168.43.186	192.168.43.1	TCP	58	36954 > 3306 [	Normal	No Attack	
18	17	0.438592	192.168.43.1	192.168.43.186	TCP	54	1720 > 36954 [	Attack	Nmap Port Scan	

Figure 4.1 ECU IoHT Dataset

WUSTL-EHMS-2020

The WUSTL-EHMS-2020 dataset was collected from a real-time Enhanced Healthcare Monitoring System testbed at Washington University in St. Louis, in which physiological sensors attached to a patient stream biometric data through a gateway to a monitoring server. The dataset contains 16,318 labelled records, of which 14,272 (87.5%) are normal and 2,046 (12.5%) correspond to man-in-themiddle spoofing or data-injection attacks, described by 35 network-flow features and 8 biometric features. This dataset is used to evaluate the system's ability to combine network-level and biometric-level evidence, which prior work has shown improves detection accuracy over using either feature type alone.



	A	B	C	D	E	F	H	I	J	K	L	M	N	O	P	Q	R	S
1	Dir	Flgs	SrcAddr	DstAddr	Sport	Dport	SrcBytes	DstBytes	SrcLoad	DstLoad	SrcGap	DstGap	SIntPkt	DIntPkt	SIntPktAct	DIntPktAc	SrcJitter	DstJitter	sMaxPktSz
2	->	e	10.0.1.172	10.0.1.150	58059	1111	496	186	276914	92305	0	0	3.582333	1.9015	0	0	2.946239	1.6235	310
3	->	e	10.0.1.172	10.0.1.150	58062	1111	496	186	230984	76995	0	0	4.294667	2.9015	0	0	3.091654	2.8625	310
4	->	e	10.0.1.172	10.0.1.150	58065	1111	496	186	218470	72823	0	0	4.540667	3.2945	0	0	2.849841	3.1655	310
5	->	e	10.0.1.172	10.0.1.150	58067	1111	496	186	203376	67792	0	0	4.877667	3.332	0	0	2.452252	3.257	310
6	->	e	10.0.1.172	10.0.1.150	58069	1111	496	186	235723	78574	0	0	4.208333	2.8635	0	0	3.021835	2.8225	310
7	->	e	10.0.1.172	10.0.1.150	58074	1111	496	186	191284	63761	0	0	5.186	3.803	0	0	3.666978	3.797	310
8	->	e	10.0.1.172	10.0.1.150	58076	1111	496	186	217958	72653	0	0	4.551333	2.841	0	0	3.360778	2.822	310
9	->	e	10.0.1.172	10.0.1.150	58078	1111	496	186	244536	81512	0	0	4.056667	2.139	0	0	3.272284	2.018	310
10	->	e	10.0.1.172	10.0.1.150	58081	1111	496	186	203989	67996	0	0	4.863	4.17	0	0	1.137211	4.066	310
11	->	e	10.0.1.172	10.0.1.150	58084	1111	496	186	235518	78506	0	0	4.212	2.319	0	0	3.300342	2.275	310
12	->	e	10.0.1.172	10.0.1.150	58086	1111	496	186	283402	94467	0	0	3.500333	2.8505	0	0	2.491458	2.8065	310
13	->	e	10.0.1.172	10.0.1.150	58087	1111	496	186	141103	47034	0	0	7.030333	2.7055	0	0	6.544402	2.6715	310
14	->	e	10.0.1.172	10.0.1.150	58089	1111	496	186	235723	78574	0	0	4.208333	2.8555	0	0	2.842504	2.8015	310
15	->	e	10.0.1.172	10.0.1.150	58094	1111	496	186	256817	85606	0	0	3.862667	2.834	0	0	2.734343	2.736	310
16	->	e	10.0.1.172	10.0.1.150	58095	1111	496	186	256618	85539	0	0	3.865667	2.859	0	0	2.606134	2.672	310
17	->	e	10.0.1.172	10.0.1.150	58097	1111	496	186	218422	72807	0	0	4.541667	3.318	0	0	2.194388	3.298	310
18	->	e	10.0.1.172	10.0.1.150	58103	1111	496	186	240679	80226	0	0	4.121667	2.345	0	0	3.183501	2.31	310

Figure 4.2 WUSTL-EHMS Dataset

Table 4.1 Summary of Datasets Used for Training and Evaluation

Dataset	Source	Samples	Feature Composition	Attack Types
ECU-IoHT	Edith Cowan University	111,207	Network + device-level	Multiple device- and network-level exploits
WUSTL-EHMS-2020	Washington University in St. Louis	16,318	35 network-flow + 8 biometric features	MITM spoofing, data injection

To emulate a realistic multi-hospital deployment during evaluation, each dataset will be partitioned across a set of simulated Flower clients. Both identical (IID) and non-identical (non-IID) partitioning strategies are planned, since hospitals in practice will not observe the same attack mix or the same volume of traffic, and a system that performs well only under an IID assumption would not reflect real deployment conditions.

4.2 DATA PREPROCESSING

Raw traffic captured at the edge gateway is normalized, checked for missing values, and anonymized before it is used for local training. Categorical protocol fields are encoded numerically, continuous network-flow and biometric features are scaled, and any field that could directly identify a patient or device owner is removed or hashed prior to feature extraction, so that only de-identified, modelready feature vectors are retained on site.

4.2.1 Algorithms

1. Local and global classification:

   XGBoost is used for both the local, perhospital classifiers and the aggregated global classifier. Gradient-boosted decision trees were selected over deep neural

architectures because they train quickly on tabular network-flow and biometric features, require comparatively little hyperparameter tuning, and produce feature-attribution values that a SHAP TreeExplainer can compute efficiently on edge-class hardware.

2. Federated aggregation:

The Flower framework coordinates communication between the hospital clients and the central server. At the start of training round t, the server broadcasts the current global parameters to every participating client. Each client k trains locally for a fixed number of epochs on its own dataset of size n , without exchanging raw samples with the server or with other clients, producing an updated local parameter set. Clients upload only this parameter set back to the server, which aggregates the received updates using Federated Averaging:

a.

$$w^{(t+1)} = \sum_{k=1}^{K} \frac{n_k}{N} w_k^{(t)}, \qquad N = \sum_{k=1}^{K} n_k$$

where K is the number of hospitals participating in round t and N is the total number of local training samples across all of them. The resulting global model is broadcast back to all clients, and the process repeats for a fixed number of rounds or until validation performance on a held-out set stops improving.

3. Risk-adaptive scoring:

The output of the global (or local) classifier is converted into an attack-probability score using the model's predicted class probability. A device trust score is maintained per device based on its recent alert history, and the two are combined into a discrete risk level (for example, Low, Medium, or High), allowing the dashboard to prioritize a small number of high risk events instead of presenting every anomaly with equal urgency.

4. Explainability:

SHAP is applied through its TreeExplainer implementation, which is well suited to gradient-boosted tree models such as XGBoost and can compute exact feature attributions efficiently enough to explain alerts in near real time. For every alert, the top contributing network-flow or biometric features are surfaced alongside the risk score.

5. Audit ledger:

The blockchain audit layer is implemented as a SHA-256 hashchained ledger: each new block stores the hash of the alert or model-update record together with the hash of the preceding block, so that altering any historical entry would be detectable through a chain-verification pass. This lightweight, application-level ledger can be operated as-is for a single organization deployment or migrated onto a permissioned distributed ledger platform if multiple independently operated hospitals need to share the same audit trail without a central operator.

6. Implementation stack:

   The dashboard is built with React.js; the backend service layer uses FastAPI for asynchronous REST and WebSocket endpoints; and PostgreSQL stores the device registry, alert and log records, federated model-version metadata, and references to the corresponding audit-hash entries.

4.2.2 Results and Discussion

At the current stage of the project, the system architecture, data pipeline, and federated training workflow described above have been designed and are being implemented; end-to-end training runs on the two datasets are in progress and have not yet been completed. This section therefore describes the evaluation methodology that will be used to assess the system, together with performance benchmarks reported by closely related architectures in prior work, which serve as reference targets rather than results obtained from the present implementation.

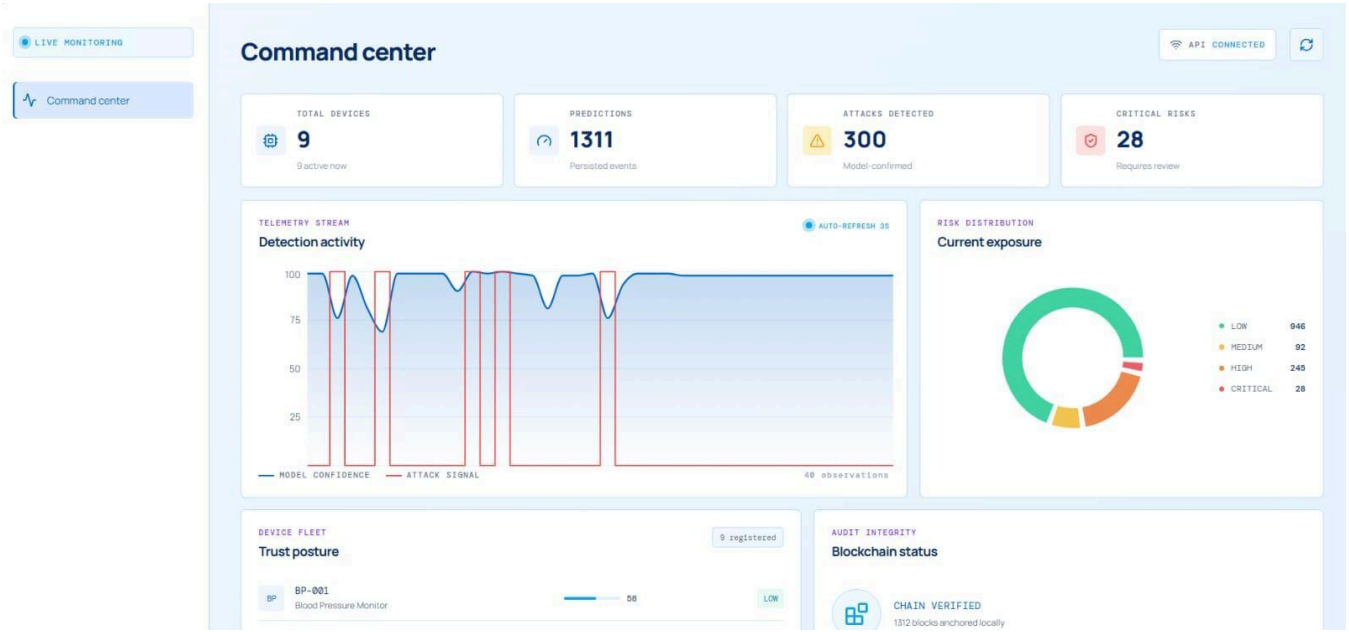


Figure 4.3 Dashboard Command Center

The global and local XGBoost models will be evaluated using accuracy, precision, recall, F1-score, and ROC-AUC on held-out partitions of both datasets. Federated-learning-specific metrics will include the number of communication rounds required to reach a target validation accuracy and the total volume of data exchanged between clients and server, since these directly affect deployability on hospital networks with limited bandwidth. Detection latency will be measured end-to-end, from feature extraction at the edge gateway to alert generation at the dashboard, and block-commit and chain-verification time will be measured to confirm that the audit mechanism does not introduce a bottleneck in the alerting path. Beyond quantitative

accuracy, the SHAP explanations produced for a sample of true-positive and false-positive alerts will be reviewed to check that the top-ranked features are clinically and technically plausible.

Prior studies that apply XGBoost and SHAP to the WUSTL-EHMS-2020 dataset in a centralized setting report accuracy in the high 90s percent range together with near-perfect ROC-AUC, and studies that combine federated learning with XGBoost and SHAP on comparable IoT intrusion datasets report accuracy consistently above 99% [3]. Federated, blockchain-audited healthcare IDS architectures evaluated on IoT healthcare traffic have similarly reported accuracy above 99%, together with measured blockchain overhead low enough for real-time operation [4], [5]. These figures are cited only as evidence that the individual techniques combined in this design are each capable of strong performance in isolation; they are not claimed as results of the present implementation, and the actual performance of the federated system described in this report will be reported once training and evaluation on ECU-IoHT and WUSTL-EHMS-2020 are complete.

CHAPTER 5

CONCLUSION

This project presented the design of a federated, explainable, and blockchain audited intrusion detection system for the Internet of Medical Things. By keeping IoMT traffic and patient data inside each hospital and exchanging only XGBoost model weights through a Flower-coordinated Federated Averaging process, the system avoids the central data-pooling assumption that most existing IoMT-IDS proposals still rely on. A risk-adaptive scoring layer turns raw predictions into prioritized, actionable alerts, SHAP-based explanations give hospital security teams a transparent basis for every alert, and a SHA-256 hashchained audit layer makes the resulting alert and model-update history tamperevident. The complete pipeline is exposed through a React.js dashboard backed by a FastAPI service layer and a PostgreSQL database, and it is designed to be evaluated on the ECU-IoHT and WUSTL-EHMS-2020 datasets, which together cover device-level and combined network-biometric attack scenarios relevant to real hospital deployments.

The system design was guided throughout by a clear threat model and an explicit mapping of security controls to identified threats, ensuring that confidentiality, integrity, and availability considerations were built into the architecture rather than added afterward. This distinguishes the proposed system from many existing IoMT-IDS proposals, which treat federated learning, explainability, and audit integrity as separate, isolated features rather than as parts of a single coherent security design.

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